Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/31/2006
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Combustion of Solid Fuel in Very Low Speed Oxygen Streams		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/1998	**End Date:**	03/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sacksteder, Kurt R (NASA Glenn Research Center) Ferkul, Paul V (National Center for Microgravity Research)		
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Performance Goal Text:			
Task Description:	This task consists of the development and flight of a space-based experiment, the continued pursuit of ground-based results in drop towers and aircraft experiments, and the continued development of a comprehensive predictive model of flame spread and extinction over solid fuels. The overall objectives of the research program are to understand the flame spread and extinction phenomena of a thin combusting solid in low-speed forced concurrent flow. Specifically, we plan to carryout a space experiment: (1) to determine the low-speed flammability boundary of a selected thin solid fuel or fuels. To verify the existence of and quantify the reasons for the critical low oxygen limit, (2) to measure the flame spread rate, flame size, and selected specific aspects of flame structure in low-speed flow, and (3) together with model predictions, determine the mechanisms controlling the behavior of flame spreading and extinction of thin solids in low-speed forced flow. In normal gravity, buoyant air motion in flames is at least 20 to 30 cm/sec. Flame spreading mechanisms that are present only in lower flow velocities, such as spacecraft air-ventilation currents, cannot, therefore,		

	be studied in normal gravity. This research seeks to contribute to fundamental combustion science, and to improve the basis for engineering improved spacecraft fire safety.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Modeling (1) A Computational Study of Low Oxygen Flammability for Thick Solid Samples The LOI apparatus is a standard device commonly used for numerical flammability ranking of the solids. The standard specimens used in the test are thick (in the form of rods and bars). The specimens are ignited at the top in normal gravity (directed downwards) and imposed forced flow of known mixture of oxygen and nitrogen. As oxygen level in the flowing gas mixture is reduced, the flame, which is normally anchored on the sides of the fuel, retreats to the wake region behind the flattened top end of the specimen (if given sufficient experimental time). The flame in the wake eventually extinguishes on further reduction in oxygen level. Here in this work, a two-dimensional numerical model is used to study the flame stabilization and extinction characteristics over a thick solid fuel. In a mixed buoyant-forced flow environment considered here, it is observed that at higher flow velocities or gravity and at low oxygen levels the flame stabilization point shifts from the sides of the wick to the wake region. The structural details in the two modes of flame stabilization are discussed. The shift in the flame anchoring position with oxygen level exhibits a hysteresis, i.e. multiple solutions may exist for a given environmental condition. The flame extinction limits and flame-shift limits (form sides of the solid fuel to the wake region or from the wake region to the sides of the fuel) in oxygen percentage are presented for the environmental parameters, forced flow velocity and gravity level. (2) Effects of Fuel Lewis Number on the Flame Spread over Solids Using a detailed two-dimensional numerical model, a systematic investigation has been made to study the effect of fuel Lewis number (LeF = ? / DF) and mass transfer on flame spread over thin solids. The fuel Lewis number affects the flame spread rates for both concurrent and opposed flames over thin fuels. The dependence of the flame spread rate on fuel Lewis number for these two spreading modes are, however, not the same. In opposed flame spreads (zero-gravity, self-propagation and normal gravity downward propagation), the flame spread rate vs. the fuel Lewis number curve is non-monotonic. In the high fuel Lewis number range, increasing fuel Lewis number decreases the opposing spread rate whereas in the low fuel Lewis number range, decreasing fuel Lewis number decreases the opposing spread rate. In steady, concurrent spread in zero-gravity with low-speed flow and a constant flame length, the flame spread rate decreases with fuel Lewis number in a monotonic manner. By using the computational model as a tool, the effects of fuel mass diffusion perpendicular to and parallel with the solid surface are isolated to obtain more physical insight on the two-dimensional aspect of fuel mass transfer on flame spread. In addition, the model has also been used to decouple the solid evaporation process so that the fuel diffusion effect in the gas phase can be isolated. Both of these theoretical exercises contribute to the understanding of mass transfer effects on the flame spreading phenomena over solids. Experimental (1) Solid Combustion Experiments Using a Continuous Fuel Dispenser for Use in Microgravity and Associated Model Computations The conventional way of determining the flammability characteristics of a material involves a number of tedious single-sample tests to distinguish flammable from non-flammable conditions. A novel test device and fuel configuration has been developed that permits multiple successive tests for indefinite lengths of thin solid materials. In this device, a spreading flame can be established and held at a fixed location in front of optimized diagnostics while continuous variations of test parameters are made. This device is especially well-suited to conducting experiments in space (e.g. aboard the International Space Station) where the limited resources of stowage, volume, and crew time pose major constraints. A prototype version of this device was tested successfully in both a normal gravity laboratory and during low-gravity aircraft trials. As part of this ongoing study of material flammability behavior, a numerical model of concurrent-flow flame spread is used to simulate the flame. Two and three-dimensional steady-state forms of the compressible Navier-Stokes equations with chemical reactions and gas and solid radiation are solved. The model is used to assist in the design of the test apparatus and to interpret the results of microgravity experiments. This paper describes details of the fuel testing device and planned experiment diagnostics. A special fuel, developed to optimize use of the special testing device, is described. Some results of the numerical flame spread model are presented to explain the three-dimensional nature of flames spreading in concurrent flow and to show how the model is used as an experiment design tool. (2) One-sided Flame Spread Phenomena of a Thermally Thin Composite Cotton/Fiberglass Fabric As an experimental necessity, past flame spread studies have relied on fast burning cellulosic papers. For longer duration tests planned for the International Space Station a 50% fiberglass, 50% cotton composite fabric is better suited for the novel fuel feeding system in the compact hardware design of a current microgravity combustion experiment. The fabric's combustion characteristics in normal gravity include unexpected cases where a flame can be sustained on one side of the fuel. One-sided flames are smaller in size than their two-sided counterparts, and propagate at half the speed. Surface temperature distributions were measured using infrared imaging and indicated a high temperature region caused by the non-flammable fiberglass. Breaching the fiberglass matrix made it possible for the flame to transfer to the other side of the fuel, suggesting that the fiberglass matrix acts as a flame arrester.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Kumar A, Tolejko KD, T'ien JS. "A computational study on flame-solid radiative interaction in flame spread over thin solid-fuel." Journal of Heat Transfer 2004 Aug;126(4):611-20. http://dx.doi.org/10.1115/1.1773196 , Aug-2004
Articles in Peer-reviewed Journals	Ferkul PV, Kleinhenz J, Shih H-Y, Pettegrew R, Sacksteder K, T'ien J. "Solid fuel combustion experiments in microgravity using a continuous fuel dispenser and related numerical simulations." Microgravity Science and Technology 2004 Jun;15(2):3-12. http://dx.doi.org/10.1007/BF02870953 , Jun-2004

Articles in Peer-reviewed Journals	Tolejko K, Feier I,I, T'ien JS. "Effects of fuel Lewis number on flame spread over solids." Proceedings of the Combustion Institute 2005 Jan;30(2):2263-70. http://dx.doi.org/10.1016/j.proci.2004.08.129 , Jan-2005